Pristyn Care Health — Patient Funnel Analysis

Milestone 1: Stakeholder Brief & Data Inventory | Yashwanth J V Nayak · Data Analyst Intern · Cynaris Solutions

1. Five Analytical Questions

1. What is the actual conversion / repeat-purchase rate by acquisition channel (Marketplace, App, Website, Offline), and which channel retains customers best?
2. Where in the customer lifecycle (first purchase → repeat orders → churn) is the biggest drop-off, based on order count and churn flags?
3. Which product categories drive the highest revenue vs. the highest return rate, and does this vary by city or channel?
4. Does discount level correlate with return rate, suggesting discount-driven low-intent purchases?
5. What is the average time-to-repeat-purchase, and can it serve as an early churn-risk signal?

2. Data Inventory

File	Rows	Cols	Missing	Date Range	Key Notes
da_sales_transactions.csv	1,000	11	0	Jul 2024 – Jun 2026	Transaction-level: units, price, discount, channel, returns
da_customer_data.csv	200	10	0	Jul 2024 – May 2026	Customer master: channel, orders, revenue, churn flag
da_product_catalogue.csv	50	7	0	—	Product master: MRP, COGS, stock, avg. monthly sales

Referential integrity: every Customer_ID and Product_ID in the transactions file matches its master table — no orphan records. Zero duplicate Transaction_IDs.

3. Completeness & Reliability Assessment

Field-level quality is high: no missing values across any of the three files, no duplicate transaction records, and clean joins between transactions, customers, and products.

Category mismatch: the transactions file references 8 product categories (including Sports, Books, Health) while the product catalogue lists only 5 (Beauty, Home, Electronics, Apparel, Grocery) — roughly 3 categories' worth of transactions cannot be joined to margin/MRP data.

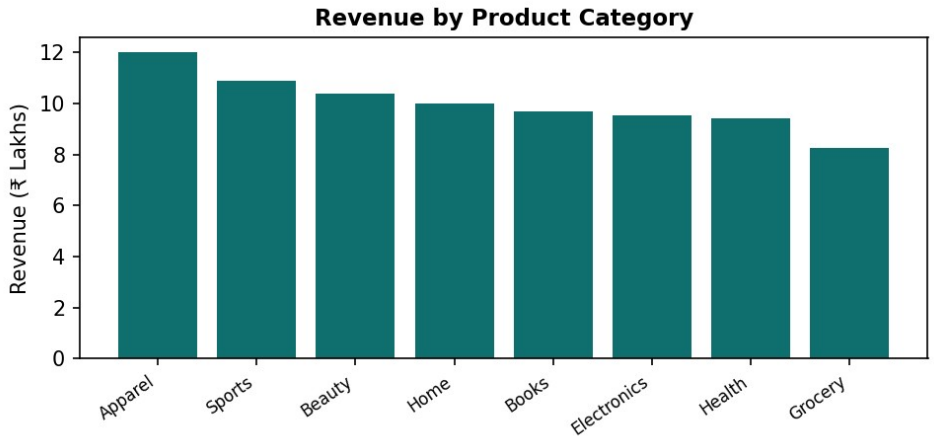
Churn under-representation: only 6 of 200 customers (3%) are flagged as churned, which is low for a healthcare/elective-service context and likely understates real attrition.

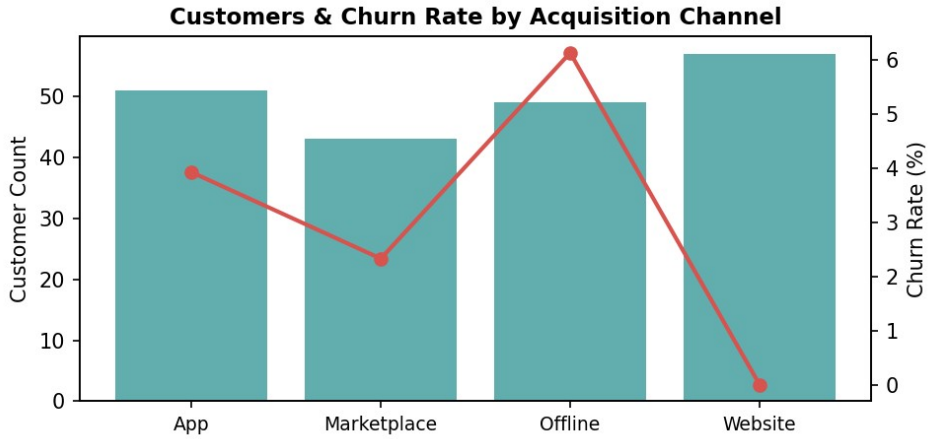
Critical Finding: Data – Brief Mismatch

The three provided files are generic multi-category retail transaction data. None contain fields for the 7-stage patient funnel described in the brief (ad click → enquiry → consultation → diagnostics → booking → surgery → follow-up). There is no top-of-funnel data (impressions, clicks, enquiries) — only completed transactions and existing customers. As-is, this supports a post-purchase retention/churn analysis, not a pre-purchase conversion funnel. Recommend confirming with the stakeholder whether funnel-stage data exists separately, or whether the funnel should be reframed around available signals (acquisition → first purchase → repeat orders → churn) before Checkpoint 2.

4. Exploratory Snapshot (Available Data)

Preliminary cuts of the available transaction and customer data, to be extended once funnel-stage data is confirmed:





Website-acquired customers show 0% churn in this sample versus 6.1% for Offline — a signal worth validating once sample size grows. Apparel and Sports lead category revenue; Grocery has the highest return rate (13.7%).

5. Next Steps into Checkpoint 2

- Escalate the funnel-stage data gap to the stakeholder/mentor for clarification.
- Pending clarification, build the data pipeline around the confirmed proxy funnel (acquisition → first purchase → repeat orders → churn).
- Set up version-controlled analysis scripts (Git) and a repeatable SQL/Python pipeline per Cynaris coding standards.